AI Model Inversion Attack & Defense

Privacy Vulnerabilities in Facial Recognition Models

P009 — Security and Privacy in Machine Learning

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Can private training faces be reconstructed using only model weights?

Roadmap

1. **Problem** — Why trained models leak private data
2. **Dataset** — ORL Faces benchmark
3. **Vulnerable Model** — Designed for maximum memorisation
4. **Attack** — Gradient ascent reconstruction
5. **Defense** — Five-layer privacy shield
6. **Evaluation** — Before vs. after, with formal guarantees
7. **Conclusion** — Key findings and takeaways

The Privacy Problem

The Situation

Modern AI systems are often shared publicly.
Trained models may **unintentionally store private information**.

The Question

Can an attacker recover private training data directly from the model?

Our Focus

Face recognition models trained on sensitive biometric data.

EU GDPR, US HIPAA, and China's PIPL all require that trained models do not leak personal data. This project demonstrates exactly when they do.

What is Model Inversion?

Normal direction:

$$\text{Face} \longrightarrow \text{Model} \longrightarrow \text{Identity}$$

Attack direction:

$$\text{Model} \longrightarrow \text{Reconstructed Face}$$

Key Idea

The model itself becomes a storage system for sensitive information.
An attacker with only the model weights can work **backwards** to recover training faces.

Dataset: ORL Faces

- **40** subjects
- **10** images per subject
- **400** total images
- Grayscale (PGM format)
- Resized to 64×64 pixels

Train / Test Split:

- 5 images per subject $\rightarrow$ training (200 total)
- 5 images per subject $\rightarrow$ testing (200 total)

Why Vulnerable?

With only **5 training images per person**, the model cannot learn general face patterns. It instead **memorises** the exact pixel patterns it saw.

Memorisation is precisely what makes inversion attacks possible.

Building the Vulnerable Model

Architecture (HighlyVulnerableCNN):

- Conv: $1 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels
- `AvgPool2d` (preserves detail)
- FC: $16{,}384 \rightarrow 1{,}024 \rightarrow 40$
- LeakyReLU ($\alpha = 0.1$) activations

Deliberate weaknesses:

- $\approx$**17.2 million** parameters
- Only **200** training images
- No dropout
- No weight decay

Goal

Create a **worst-case privacy scenario**.

Result

100% training accuracy = **perfect memorisation**.

Why AvgPool not MaxPool?

Average pooling preserves blended pixel information. Max pooling discards $\approx$75% of activations. AvgPool makes inversion significantly easier.

Training Results

	Value	**Interpretation**
Train accuracy	**100.0%**	Perfect memorisation confirmed
Validation accuracy	**94.5%**	Still generalises well
Chance level	2.5%	40-class random baseline

Interpretation

The model memorised every training image **and** still performs on unseen faces. This combination — high accuracy *and* high memorisation — is the ideal environment for a successful inversion attack.

How the Attack Works

Start: Random Gaussian noise — no face, no information.

Optimisation loop (gradient ascent, 500 iterations):

$$x_{t+1} = x_t + \eta \, \nabla_x \log P(y \mid x)$$

- $\eta = 0.05$ (learning rate)
- At each step: ask the model "what pixel change increases confidence for Subject k?"
- Update x in that direction

Goal

Find the image x^* that the model believes belongs to Subject k with maximum confidence.

Making the Reconstruction Realistic

Problem: Unconstrained optimisation produces noisy, adversarial-like artifacts.

Solution: Total Variation (TV) regularisation encourages spatial smoothness.

$$\mathcal{L}_{\text{total}} = -\log P(y \mid x) + \lambda_{\text{TV}} \cdot \text{TV}(x)$$

where $\text{TV}(x) = \sum_{i,j} \sqrt{(x_{i+1,j} - x_{i,j})^2 + (x_{i,j+1} - x_{i,j})^2}$

Effect

Neighbouring pixels are encouraged to stay smooth $\Rightarrow$ structured, face-like reconstructions instead of random noise patterns.

Hyperparameter Search

Grid search: 3 learning rates $\times$ 3 TV values $=$ 9 combinations.
Measured on 5 subjects with 300 attack iterations each.

λ_{TV}	LR	Mean SSIM	**Note**
10^{-5}	0.10	0.1452	**High noise**
10^{-3}	0.01	0.2129	**Over-smoothed**
10^{-4}	0.10	0.2730	**Good balance**
$\mathbf{10^{-3}}$	**0.05**	**0.3012**	**Optimal**

Best Configuration

$\lambda_{\text{TV}} = 10^{-3}$, LR $= 0.05$, SSIM $= 0.3012$

This is $\approx$ **32**$\times$ the random-noise baseline (SSIM $= 0.0093$).

Final model confidence: $P(y \mid x^*) = \mathbf{1.0000}$

Random noise; [draw, fill=gray!40, minimum width=2cm, minimum height=2cm, rounded corners] (n1) at (4,0) Step 200

Rough structure; [draw, fill=darkred!20, minimum width=2cm, minimum height=2cm, rounded corners, text=darkgray] (n2) at (8,0) Step 500

Recognisable face; [-¿, thick] (n0) – (n1) node[midway, above] gradient ascent; [-¿, thick] (n1) – (n2)

node[midway, above] refine;

Attack Results (All 40 Subjects)

Metric	Value	Significance
Mean SSIM (attack)	**0.2458** $\pm$ 0.0617	—
Random-noise baseline	0.0093	Reference
Improvement ratio	**26.4**$\times$	Far above chance
Mean MSE (attack)	**0.0479** $\pm$ 0.0135	Low pixel error
Mean confidence	**1.0000**	Full certainty

Conclusion

The model leaked private visual identity information.
The 26.4$\times$ gain over random baseline **rules out chance** and confirms real leakage.

Note: The ablation best-case (5 subjects) reached SSIM $= 0.3012$ *(*$\approx$*32*$\times$ *baseline). The full 40-subject mean is 0.2458 (26.4*$\times$*) — the scientifically correct figure.*

Defense Strategy

Five-layer combined defense:

1. **Differential Privacy SGD** (core mechanism)
2. **MC Dropout** ($p = 0.30$, always on at inference)
3. **Label Smoothing** ($\varepsilon = 0.10$)
4. **Batch Normalisation** (after each conv layer)
5. **Data Augmentation** (random flips, crops)

Goal

Reduce memorisation while preserving classification utility.

Differential Privacy SGD

Two-step mechanism applied per mini-batch:

Step 1 — Clip per-sample gradients (norm bound $C = 1.0$):

$$\tilde{g}_i = g_i \cdot \min\left(1,\ \frac{C}{\|g_i\|}\right)$$

Step 2 — Add calibrated Gaussian noise ($\sigma = 0.4$):

$$\bar{g} = \frac{1}{B}\left(\sum_{i=1}^{B} \tilde{g}_i + \mathcal{N}\left(0,\ \sigma^2 C^2 \mathbf{I}\right)\right)$$

Effect

Clipping limits any single training example's influence. Noise injection makes exact memorisation mathematically impossible.

Formal Privacy Guarantee

Method: Rényi Differential Privacy (RDP) accountant.

Parameters: $\sigma = 0.4$, $C = 1.0$, batch size $= 32$, $N = 200$, 6 DP epochs.

$$(\varepsilon, \delta)\text{-DP} \quad \text{with} \quad \varepsilon = \mathbf{20.02}, \quad \delta = 2.94 \times 10^{-3}$$

Important — Honest Assessment

The widely-cited threshold for *meaningful* formal DP is $\varepsilon \leq 10$.
Our $\varepsilon = 20.02$ exceeds this threshold.

Root cause: $N = 200$ forces a high sampling rate $q = B/N = 0.16$, which inflates the privacy cost. With a larger dataset, ε would improve significantly.

The **empirical** defence works regardless — the formal guarantee is weak and must be stated honestly.

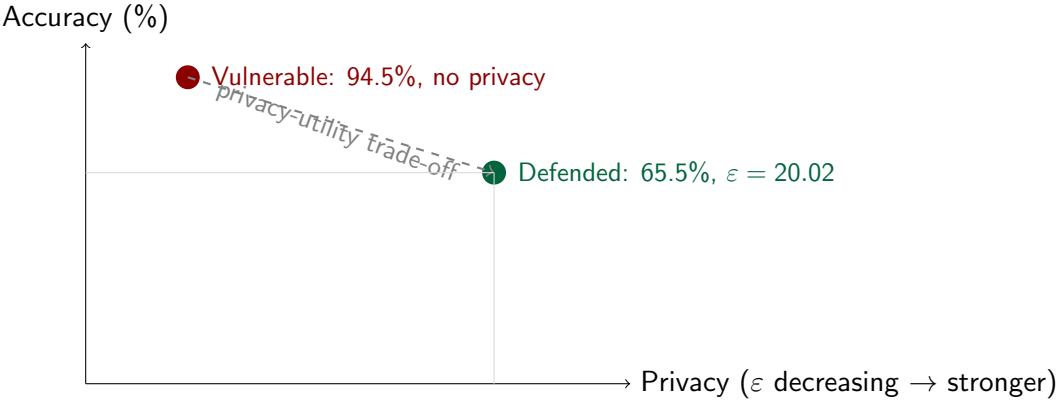
Before vs. After Defense

Metric	Vulnerable	Defended	Baseline
Val Accuracy	**94.5%**	65.5%	2.5% (chance)
Attack Confidence	1.0000	0.9171	—
Mean SSIM	0.2458	**0.0319**	0.0093
Mean MSE	0.0479	**0.0838**	0.1226
Leakage ratio	26.4×	3.4×	1×
Privacy ε	∞ (none)	**20.02**	—

Result

Attack SSIM reduced by **87%** (from 0.2458 to 0.0319).
Defended SSIM is only **+0.0226** above random noise — within statistical noise.

Privacy vs. Utility Trade-off



Trade-off

Better privacy caused a **29-percentage-point** accuracy drop. Perfect privacy and perfect accuracy cannot coexist on this dataset.

Reproducibility: 3-Seed Study

Seed	**Vulnerable SSIM**	**Defended SSIM**
42	0.1974	0.0319
7	0.1932	0.0334
123	0.1943	0.0408
Mean $\pm$ std	0.1950 ± 0.0018	0.0354 ± 0.0039

Interpretation

Small standard deviations relative to the gap between models confirm that the defence effect is **real and stable**, not an artefact of lucky initialisation.

Key Findings

1. **Model inversion is real.**
   $26.4\times$ above random baseline rules out chance.
2. **Memorisation causes leakage.**
   100% training accuracy $\Rightarrow$ the model becomes a data store.
3. **Differential privacy reduces leakage by 87%.**
   SSIM drops from 0.2458 to 0.0319, near random noise.
4. **Privacy has a measurable cost.**
   29-point accuracy penalty on this small dataset.
5. **Formal guarantees require honest reporting.**
   $\varepsilon = 20.02$ is real DP — but not strong DP. Say so.
6. **Perfect privacy and perfect accuracy cannot coexist.**

Final Takeaway

"A model that memorises is a model that leaks."

Privacy is not free. It must be **engineered into the model from the start.**

If you build AI systems on sensitive data — **medical**, **financial**, or **biometric** — privacy cannot be optional.

The Single Highest-Leverage Fix

The weak ε and accuracy drop both stem from $N = 200$. More training data is the most effective privacy improvement.

Thank you.

References

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